Due on monday, November 28th

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1. Potential scattering.

   Consider a particle of mass m on a one-dimensional line. The potential is zero to the left of $x = -a$ and has a constant positive value to the right of $x = a$. The potential has unknown features within the region $x \in [-a, a]$.

$$V(x) = \begin{cases} 0 & \text{for } x < -a \\ \text{unknown} & \text{for } -a < x < a \\ V_0 & \text{for } x > a \end{cases}$$

   The Schroedinger equation for this system is found/known to have a solution with energy E with the following forms in the regions outside $[-a, a]$:

$$\begin{aligned} \psi(x) &= \alpha e^{ik_1 x} + \beta e^{-ik_1 x} && \text{for } x < -a \\ \psi(x) &= \gamma e^{ik_2 x} && \text{for } x > a \end{aligned}$$

   where

$$\alpha = \frac{k_1 + ik_2}{k_1 - ik_2} \quad ; \quad \beta = \frac{ik_1}{k_1 - ik_2} \quad ; \quad \gamma = \frac{\sqrt{k_1 k_2}}{k_1 - ik_2}$$

   (a) **[4 pts.]** Express k_1 in terms of E.

   (b) **[4 pts.]** Is E larger or smaller than V_0? Explain why.

   (c) **[5 pts.]** Express k_2 in terms of E and V_0.

   (d) **[3 pts.]** If we want to interpret this as a scattering problem, which parts of the wavefunction would you interpret as incident wave, reflected wave, and transmitted wave?

   (e) **[4 pts.]** Calculate the reflection coefficient.

   (f) **[5 pts.]** Calculate the transmission coefficient.
   (Careful: remember momentum factors.)

   (g) **[3 pts.]** Check that the sum of the two coefficients makes sense.

2. Suppose $\psi_0(x)$ is a normalized wavefunction with

$$\langle \hat{x} \rangle_{\psi_0} = \langle \psi_0 | \hat{x} | \psi_0 \rangle = x_0 \qquad \langle \hat{p} \rangle_{\psi_0} = \langle \psi_0 | \hat{p} | \psi_0 \rangle = p_0$$

where x_0 and p_0 are constants. The subscript indicates the state in which the expectation values are being calculated. Define a new wavefunction

$$\psi_{new}(x) = e^{iqx/\hbar} \psi_0(x)$$

where q is a real quantity having dimensions of momentum.

(a) **[6 pts.]** What is the expectation value $\langle \hat{x} \rangle_{\psi_{new}}$ in the state $\psi_{new}(x)$?

(b) **[7 pts.]** What is the expectation value $\langle \hat{p} \rangle_{\psi_{new}}$ in the state $\psi_{new}(x)$?

(c) **[2 pts.]** Based on your results, interpret in one sentence the physical significance of multiplying a wavefunction by the factor $e^{iqx/\hbar}$.

3. Consider a particle with mass m in the state described by wavefunction $\psi(x)$, which we write in amplitude-phase form:

$$\psi(x) = A(x)e^{i\theta(x)}$$

where both $A(x)$ and $\theta(x)$ are real functions.

(a) **[SELF]** Write down the definition of the probability current density.

(b) **[5 pts.]** Show that the probability current density is given by

$$j(x) = [A(x)]^2 \, \frac{\hbar}{m} \, \partial_x \theta(x)$$

i.e., the probability current is proportional to the *gradient* of the phase.

(c) **[2 pts.]** Show that there is no current in a region where the wavfunction is real.